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Software Engineering

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Lab 1

Problem statement: 11

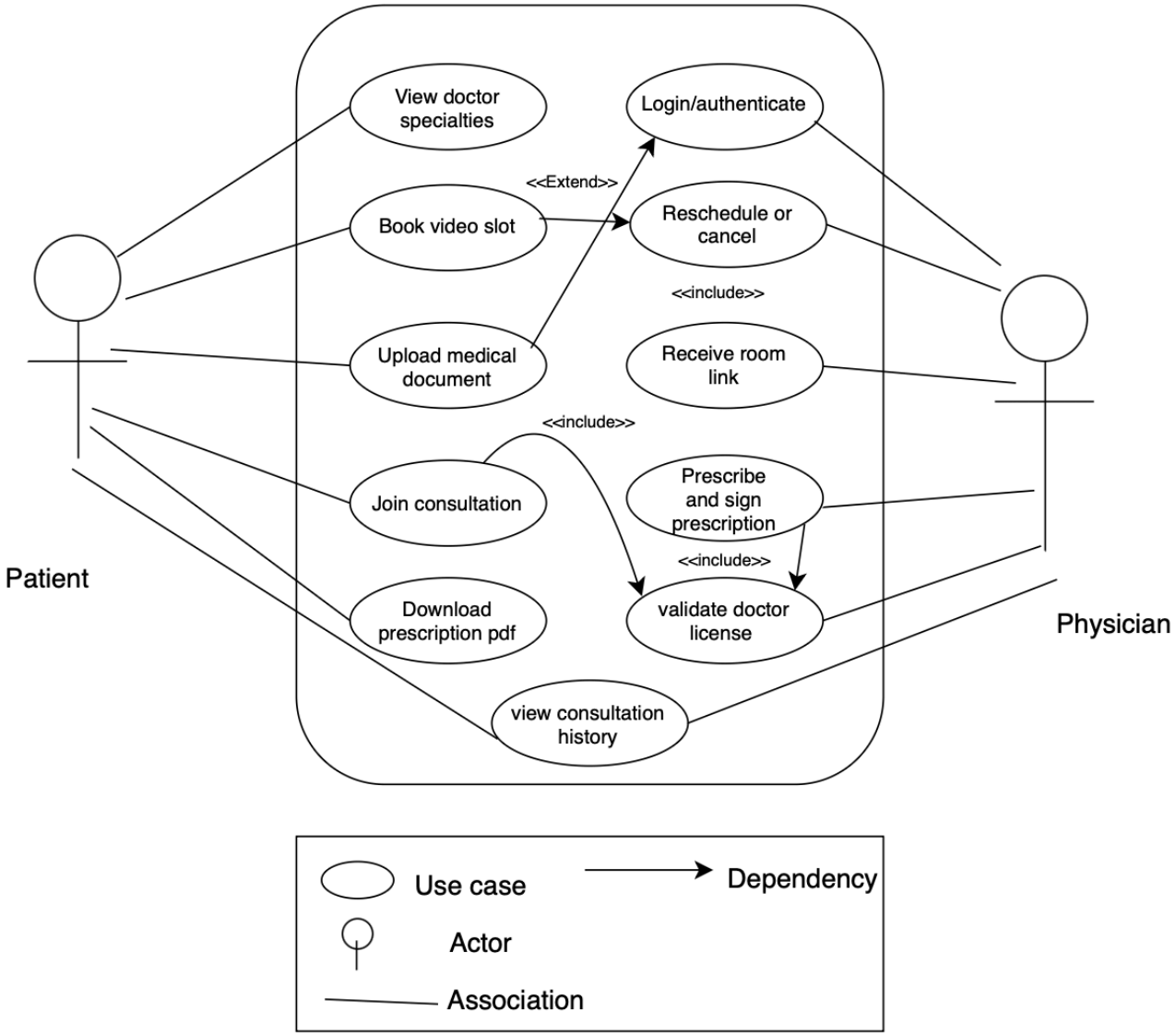
Deliverable 1:

Id	Type	Description	Priority	Acceptance rate	Rationale
FR-001	Functional	The system must allow the users to view the doctor's speciality, availability, reputation etc	High	Pass: Search results display only those doctors matching the selected specialty and showing real time available slots. Fail: Booked/ unavailable slots appear as selectable.	Ensures that patients can make better decisions about who they want treating them before booking the slot.
FR-002	Functional	The system must allow physicians to prescribe and digitally sign prescriptions that are automatically converted into downloadable PDF.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures prescriptions are authentic, verifiable, and legally valid.

FR-003	Functional	The system must allow the users to upload documents secured through an encrypted database	High	Pass: An uploaded document is stored in encrypted form and can only be accessed by the patient and the physician assigned to them Fail: A document is stored in plaintext, or a user without a doctor-patient relationship to that document is able to view or download it.	Ensures that confidential documents aren't accessed by those who're not supposed to access it won't
FR-004	Functional	The system must allow patients to view and download their consultation history and prescriptions from a personal dashboard.	Medium	Pass: Dashboard lists all past consultations with a working download link for each prescription. Fail: A prescription is missing or inaccessible for a completed consultation.	Provides patients continuous access to their medical records.
FR-005	Functional	The system must generate and share an encrypted, time-bound video consultation room link with both the patient and physician prior to the scheduled slot.	High	Pass: Link becomes active only within the scheduled time window and uses end-to-end encryption. Fail: Link is accessible before/after the valid window or is unencrypted.	Provides secure, time based access to the session.

NFR-001	Non-Functional	All video session metadata and patient clinical notes must be encrypted in transit using TLS and at rest with AES-256	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Protects sensitive health data confidentiality and complies with healthcare data
NFR-002	Non-Functional	The system shall maintain 99.9% uptime for slot booking and consultation link generation services, with failover recovery within 30 seconds.	High	Pass: Load and chaos-testing logs confirm uptime greater than 99.9% over a 30-day window and failover completes within 30 seconds. Fail: Booking service downtime exceeds allowed threshold or failover takes longer than 30 seconds.	Ensures patients can reliably access time-sensitive medical consultations without service interruption.

Deliverable 2:



Deliverable 3:

Use-Case Flow Specification

Use Case Name	Book Video Consultation Slot
Use Case ID	UC-02
Primary Actor	Patient
Secondary Actor	Attending Physician, Payment Gateway (external)
Related Requirements	FR-001 (browse doctor availability), FR-005 (generates the encrypted consultation link after booking), NFR-002 (booking service reliability)
Includes	«include» Authenticate Patient is invoked at booking initiation to verify the patient's identity/session before allowing slot selection.
Extends	«extend» Reschedule Consultation Slot: Optionally triggered from this use case when a patient chooses to change an existing booking instead of creating a new one.
Preconditions	1. The patient has a registered and authenticated account on the portal. 2. At least one physician has published available consultation slots for a future date/time. 3. The patient has completed or is ready to complete the consultation fee payment.
Postconditions	1. The selected slot is marked as reserved and removed from the physician's publicly available calendar. 2. A booking confirmation is recorded in the system and sent to the patient. 3. An encrypted, time-bound video consultation room link is generated and scheduled for release to both patient and physician.

Main Success Scenario	1. Patient logs into the portal and navigates to Book Consultation. 2. System authenticates the patient session («include» Authenticate Patient). 3. Patient searches for a physician by specialty and views available time slots. 4. Patient selects a specific date and time slot from the physician's calendar. 5. System temporarily locks the selected slot to prevent concurrent booking by another patient. 6. Patient confirms the booking and completes payment via the integrated payment gateway. 7. System permanently reserves the slot, generates a booking confirmation, and records the transaction. 8. System schedules generation of an encrypted video consultation room link tied to the confirmed slot. 9. System sends booking confirmation notifications (with slot details) to both the patient and the physician.
Alternate Flow	**A1: Slot Becomes Unavailable During Booking** 1. At step 5, the system detects that another patient has already confirmed the same slot before the current patient completes payment. 2. System cancels the temporary lock for the current patient and displays a "Slot No Longer Available" message. 3. System refreshes and displays the physician's updated list of currently available slots. 4. Patient selects an alternative slot and the flow resumes from step 4 of the Main Success Scenario.
Exception Flow	1. Payment failure: system releases the slot lock and prompts the patient to retry payment or select a different slot.